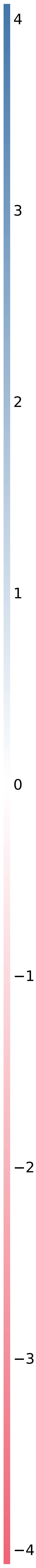
Question:

Jen enters a lottery by picking 4 distinct numbers from $S=\{1,2,3,\cdots,9,10\}.$ 4 numbers are randomly chosen from $S.$ She wins a prize if at least two of her numbers were 2 of the randomly chosen numbers, and wins the grand prize if all four of her numbers were the randomly chosen numbers. The probability of her winning the grand prize given that she won a prize is $\tfrac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Reasoning + Answer:



<think>
1. **Count all possible combinations of the winning numbers.**
There are $\binom{10}{4}$ ways to choose 4 numbers from a set of 10 numbers.
$$
\binom{10}{4} = 210
$$

2. **Determine the number of ways Jen can win the grand prize.**
For Jen to win the grand prize, she has to choose the very same 4 numbers that were randomly chosen. There is only 1 way to do this (choosing the 4 numbers together).
$$
N_{\text{grand}} = 1
$$

3. **Determine the number of ways Jen can win a prize.**
For Jen to win a prize, she has to have at least 2 of her 4 chosen numbers in the set of the randomly chosen 4 numbers. We need to sum the cases where:
- She has exactly 2 numbers (out of her 4) in the randomly chosen 4.
- She has exactly 3 numbers (out of her 4) in the randomly chosen 4.
- She has all 4 numbers in the randomly chosen 4.
We'll compute these three cases separately.

Case 1: Exactly 2 numbers are the same.
\[
\text{ Choosing 2 of the 4 chosen numbers to be the ones picked together and 2 more, one from the remaining set} \\
\binom{4}{2} \times \binom{6}{2} = 6 \times 15 = 90
\]

Case 2: Exactly 3 numbers are the same.
\[
\text{ Choose 3 chosen numbers to be the ones picked together} \\
\binom{4}{3} \times \binom{6}{1} = 4 \times 6 = 24
\]

Case 3: 4 numbers are the same.
\[
\text{ Total ways to choose the same 4 numbers} \\
\binom{4}{4} \times \binom{6}{0} = 1 \times 1 = 1
\]

Summing the cases for winning a prize.
\[
N_{\text{prize}} = 90 + 24 + 1 = 115
\]

4. **Calculate the ratio of grand prize winning probability to prize-winning probability.**
P(win grand prize) is $N_{\text{grand}} / N_{\text{win}}$. Given the required simplification to $\frac{m}{n}$.

Hence,
\[
\frac{N_{\text{grand}}}{N_{\text{prize}}} = \frac{1}{115}
\]

5. Putting both parameters back, the required m, n are integers:
\[
\frac{1}{115}.
\]
Thus, m and n are 1 and 115 respectively.
6. Compute m+n:
\[
m+n = 1 + 115 = 116
\]
</think>
<answer>116</answer>